

# Vedant Singh

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📍 Chennai, India 🔗 Portfolio

## PROFESSIONAL SUMMARY

**Microsoft Certified Azure AI Associate** and BTech student at VIT Chennai (**CGPA 9.14**) specializing in AI-integrated full-stack development. Proven technical problem-solver with a 2nd rank in the NetSim Hackathon, dedicated to bridging the gap between advanced Generative AI models and scalable, user-centric web applications.

## EDUCATION

**Btech in Electronics and Computer Engineering** August 2023 – Present  
*VIT Chennai*  
**CGPA 9.14**

Class 12, Percentage: 89.8% 2023

Class 10, Percentage: 95.8% 2021

## SKILLS

### Languages

Python, C++, Java, JavaScript, SQL, HTML/CSS.

### AI & Data Science

Machine Learning, Deep Learning, Pandas, NumPy, Seaborn, Scikit-learn, SHAP.

### Cloud & Tools

Microsoft Azure, Git/GitHub, AWS, Figma

### Web Framework

React

## PROJECTS

### Rockfall Prediction System

Engineered an end-to-end full-stack application to classify geological stability zones (Safe/Moderate/Unsafe) using a **Random Forest** classifier. Generated a comprehensive **synthetic dataset** simulating critical environmental factors (soil moisture, slope angle, precipitation) to overcome data scarcity and train the model. Developed a **Flask** backend to process environmental inputs and serve predictions to a responsive frontend dashboard. Implemented real-time data visualization charts to display risk trends, providing actionable safety insights for vulnerable areas.

### Cronos AI | Personalized AI Chatbot

Developed a hybrid voice assistant in **Python** combining low-latency offline hotword detection (**Porcupine**) with cloud-based LLM intelligence (**Gemini API**). Integrated **Google Speech Recognition** to parse voice commands, enabling system-level automation (opening apps/sites) and complex natural language Q&A.

### Diabetes Predictor using Machine Learning

Engineered a machine learning pipeline handling class imbalance with **SMOTE** and missing data via median imputation. Outperformed Logistic Regression (71%) with a **Random Forest** model (77%), validating non-linear modeling for health data. Conducted a comparative XAI study proving **LIME's** mathematical instability; implemented **SHAP** to derive robust, game-theoretic explanations for top drivers (Glucose, BMI).

## CERTIFICATIONS & ACHIEVEMENTS

### Microsoft Azure AI Associate

Certified by Microsoft, validating expertise in designing and deploying AI solutions on Azure, including integrating AI services for diverse applications

### NetSim Hackathon Rank 2

Designed and simulated networks using NetSim, optimizing network performance parameter and secured 2nd place.